Report R8 — C150 at open boundaries: an exactly solved unitary QCA, and exactly what it leaves open

Krylov sector analysis of quantum cellular automata

QCA Sector Analysis project (Tier 1)

2026-07-26 (engine_version `1a.1`)

Abstract

C150 — the quantum elementary cellular automaton with Wolfram number 150, local symbol table (`I`, `V`, `V`, `I`), written `W150` elsewhere in this series — is the member of the 16-rule unitary family whose sector structure we can solve in closed form at open boundaries. We do so here, and then measure precisely how much of the dynamics that solution does *not* determine. Three results. (i) *Reduction.* For any unitary rule the sector partition equals the connected components of the single-flip graph $x \sim x \oplus 2^i$ over the sites at which the Hadamard fires; the brick-wall ordering drops out. Verified against the streamed engine on all 16 unitary rules, both boundary conventions, $N = 3 \dots 12$ (320 units, no mismatch). (ii) *Solution.* In the bond variables $b_j = x_j \oplus x_{j+1}$ the map is a bijection onto the even-weight strings of length $N + 1$ and the elementary move is a hard-core *hop* of one domain wall. Hence the conserved charge is the open Ising domain-wall number $Q = \sum_j (1 - Z_j Z_{j+1})/2$, the sectors are its eigenspaces, $|S_w| = \binom{N+1}{w}$ for even w, the sector count is $\lfloor (N+1)/2 \rfloor + 1$ and $D_{\max} = \max_{w\,\text{even}} \binom{N+1}{w}$. Because the sector count is *linear* in N, C150 is symmetry-resolved and not fragmented: it is the natural null model for the family. The exponential base of $D_{\max}$ is exactly 2 with an $N^{-1/2}$ prefactor, which supersedes the fitted 2.022 of R2. We push the reduction to $N = 32$ — 4.29×10^9 basis states, 21 GB, 815 s, the RAM ceiling — and the assumption-free engine as far as a one-day budget reaches (see Table 1 for which units have landed). The closed form is exact at every N computed, by either route. (iii) *What is left open.* The sector is the maximal reachable set, not the dynamics. U has an exponentially large stationary space, $\dim \mathrm{Fix}(U_w) = \binom{\lceil N/2 \rceil}{w/2}$ and $\dim \mathrm{Fix}(U) = 2^{\lceil N/2 \rceil}$ (an exact law, verified two independent ways), almost all of it *dark superpositions* that no basis state names. The spectrum of U_w is degenerate for *every* shell with $|S_w| > N + 1$ and nondegenerate for the extremal ones, so outside those extremes $\dim \mathcal{K}(x) < |S_w|$ without exception: an inner sector is never the Krylov space of one of its basis states. Under monitoring the sector *is* filled uniformly (provably); without monitoring the diagonal ensemble occupies between 6.6% and 50% of it. Finally, in the bond representation C150 is a free-fermion Hadamard wall walk dressed by exactly two diagonal signs — a maximal nearest-neighbour wall interaction and a Jordan–Wigner string on the coin — and removing either one alone leaves the many-body spectrum non-additive while removing both makes it additive to 4×10^{-15}. C150 is therefore exactly charge-resolved but genuinely interacting inside each sector.

Contents

1 Why this rule, and a naming note

Throughout this series rules are named by Wolfram number with a `W` prefix (`W150`); the present report follows the user-facing shorthand *C150* for the same object. The encoding is the standard one of Tier 1: the Wolfram number is read as a 4-tuple of local symbols indexed by the control pattern (x_{i-1}, x_{i+1}), with `I` the identity and `V` the Hadamard, so

$$150 \longmapsto (\mathtt{I}, \mathtt{V}, \mathtt{V}, \mathtt{I}), \qquad \text{i.e.} \qquad \text{Hadamard fires at site } i \iff x_{i-1} \oplus x_{i+1} = 1. \tag{1}$$

The firing condition is exactly the update of the *classical* additive elementary rule 90; C150 is the automaton that applies a Hadamard wherever ECA 90 would write a one. One cycle is the brick-wall product of a layer on the even sites followed by a layer on the odd sites, controls always read from the current bitmask. Boundaries: `obc0` fixes $x_{-1} = x_N = 0$, which is the case treated here; the ring `pbc` appears only for contrast in §4.

C150 is singled out for three reasons. It is one of the two rules (150 and 105) whose sector sizes R2 identified as binomials, so a closed form was plausible; it is the rule whose `obc0` sector sizes are used as a regression fixture against the external HSF oracle in R1; and its fitted growth exponent (2.022) sat awkwardly above the informational ceiling 2, which suggested that the fit was seeing a prefactor rather than a base. All three are settled below.

2 Reduction: the sector partition of a unitary rule is a flip graph

Theorem 1 (flip reduction)**.** *Let the rule be unitary (symbols in $\{\mathtt{I}, \mathtt{V}\}$) and let $s_i(x) = t[2x_{i-1} + x_{i+1}]$ be the symbol fired at site i when the controls read x. Then the sector partition — the weakly connected components of the one-cycle transition graph, which for a unitary rule coincide with its strongly connected components — equals the connected components of the undirected graph*

$$F: \quad x \sim x \oplus 2^i \qquad \textit{for every } i \textit{ with } s_i(x) = \mathtt{V}. \tag{2}$$

Proof. A cycle is two depth-1 half-layers. By the no-interference theorem of R1, each half-layer is — conditioned on the frozen complementary sublattice — a product of single-qubit gates acting on disjoint qubits, so the image of a basis state x under one half-layer is exactly the *subcube* spanned by the sites whose symbol is `V`, with the complementary sublattice held fixed. Two consequences. First, x lies in its own half-layer image, because $\langle b|\mathtt{I}|b\rangle = 1$ and $\langle b|H|b\rangle = \pm 1/\sqrt{2}$ are both nonzero; hence the composite image $\Phi(x)$ contains both half-layer images $A(x)$ and $B(x)$. Second, inside a half-layer image the frozen sublattice — and therefore the `V`/`I` pattern itself — does not change, so the subcube is connected by single flips which are themselves edges of F. By the first point every F-edge is realised by a composite path; by the second, every composite edge $x \to z$ (with $z \in B(y)$, $y \in A(x)$) is realised by the F-path $x \sim y \sim z$. The two graphs have the same components. $\square$

Two remarks. The reduction is *layer-agnostic*: the brick-wall ordering disappears, because the controls of site i are read off x itself in the half-layer in which i is frozen. The sector partition of a unitary QECA is therefore a property of the local symbol table alone — not of the circuit schedule. And the cost collapses from $O(2^N|\mathrm{succ}|)$, with $|\mathrm{succ}|$ growing exponentially for the Hadamard-rich rules, to $O(N2^N)$ with a tiny constant; §5 uses this.

Validation. Theorem 1 is checked against the streamed engine (`graph.scc.analyze`, which knows nothing about it) for all 16 unitary rules, both boundary conventions and $N = 3 \dots 12$: 320 units, full size multisets compared, no mismatch, for both the pure-python union-find and the vectorised label-propagation implementation. See `tests/test_flip_graph.py`.

3 Solution: walls, and the Ising charge

Introduce the bond (domain-wall) variables. For `obc0` there are $N+1$ bonds,

$$b_j = x_j \oplus x_{j+1}, \quad j = 0, \dots, N, \qquad x_{-1} = x_N = 0, \tag{3}$$

with the index convention of the code ($b_0 = x_0$, $b_N = x_{N-1}$).

Proposition 1 (bond bijection). *$x \mapsto b$ is a bijection from $\{0,1\}^N$ onto the even-weight strings of length $N+1$; the inverse is the prefix XOR $x_j = \bigoplus_{k \le j} b_k$. On the ring the same map is exactly 2-to-1 onto the even-weight strings of length N, the two preimages being x and its global spin flip.*

Both statements are verified exhaustively for $N = 2 \dots 12$ (obc0) and $N = 3 \dots 12$ (pbc).

Now translate the flip move (2). Flipping x_i toggles exactly the two bonds b_i and b_{i+1}, and by (1) the flip is allowed iff $x_{i-1} \ne x_{i+1}$, i.e. iff $b_i \ne b_{i+1}$. So the elementary move is

$$(b_i, b_{i+1}) = (1,0) \ \longleftrightarrow\ (0,1): \qquad \text{a } \textit{hard-core hop} \text{ of one domain wall.} \tag{4}$$

Creation and annihilation of walls is forbidden — the configuration $b_i = b_{i+1}$ is precisely where the symbol is `I`. Therefore:

Corollary 1 (conserved charge). *The wall number $w = \sum_j b_j$ is conserved exactly, i.e. $[U, Q] = 0$ for the strictly 2-local diagonal operator*

$$Q \;=\; \sum_{j=0}^{N} \frac{1 - Z_j Z_{j+1}}{2}, \qquad Z_{-1} = Z_N = +1, \tag{5}$$

the open Ising domain-wall number with boundary fields. (Q is diagonal, so conservation at the level of supports already gives the operator identity; support-level conservation is verified for every x, $N = 3 \dots 12$, both boundary conventions, with zero violations.)

Corollary 2 (connectivity). *Each wall shell is one sector. Hard-core hopping on a segment of $N+1$ sites is connected on every fixed-occupancy shell (bubble the walls to the leftmost packed configuration), and (4) is exactly that dynamics.*

4 Closed form, and the growth law

Combining Proposition 1 with Corollaries 1 and 2:

$$\boxed{|S_w| = \binom{N+1}{w} \ (w \text{ even}), \qquad \#\text{sectors} = \left\lfloor \frac{N+1}{2} \right\rfloor + 1, \qquad D_{\max} = \max_{w \text{ even}} \binom{N+1}{w}.} \tag{6}$$

The frozen sectors are the singletons $w = 0$ (all-zero) and, when N is odd, $w = N + 1$ (the alternating configuration); for even N there is only one. The parity bookkeeping in $D_{\max}$ is not cosmetic: for $N \equiv 1 \pmod 4$ the central binomial sits at odd w and is *not* a sector size, so $D_{\max}$ is the near-central one (e.g. $N = 17$: $\binom{18}{8} = 43758$, not $\binom{18}{9}$).

Ring, for contrast. On the ring the wall map is 2-to-1, so a shell with $0 < w < N$ lifts to one sector of size $2\binom{N}{w}$, while the frozen shells $w = 0$ and (for even N) $w = N$ each split into *two* singletons — no hop is available, and the two spin preimages are disconnected. Hence #sectors $= N/2 + 3$ for even N and $(N + 3)/2$ for odd N. Both closed forms reproduce the stored Tier-1a data at every $N = 6 \dots 20$, in both boundary conventions, size multiset for size multiset.

C150 is not fragmented. Equation (6) has a sector count that is *linear* in N and sector sizes that are full eigenspaces of the single charge Q. There is no Krylov structure beyond the $U(1)$ symmetry: C150 is *symmetry-resolved*, not fragmented. This is worth stating plainly because C150 is the rule most often quoted from this family, and it is exactly the wrong example for fragmentation. The fragmented members of the family look nothing like it — `W204` has 2^N frozen singletons, and the wall families of `W51`/`W201` carry exponentially many sectors. C150 is the family's null model.

The growth law: base exactly 2, prefactor $N^{-1/2}$. Since $\sum_{w\,\text{even}} \binom{N+1}{w} = 2^N$ and the maximum is (near-) central,

$$\frac{D_{\max}}{2^N} = 2\sqrt{\frac{2}{\pi(N+1)}}\left(1 + O(N^{-1})\right) \longrightarrow 0, \tag{7}$$

so $D_{\max}$ grows as $2^N N^{-1/2}$: the exponential base is *exactly* 2 and the sub-exponential correction is a square root. The approach is not smooth but a sawtooth (Figure 1, right), which is the parity bookkeeping above: for $N \equiv 1 \pmod 4$ the true central binomial is at odd w and $D_{\max}$ has to settle for the neighbour, which costs a factor $1 - O(N^{-1})$ and produces the dips at $N = 5, 9, 13, \dots$. A pure-exponential fit sees this sawtooth on top of the $N^{-1/2}$ decay, which is exactly the configuration that pushes an unconstrained base slightly above the true one. The value 2.022 reported for C150/`obc0` in R2's pure-exponential fit is the finite-size shadow of that prefactor, not a base above the informational ceiling; R2's companion observation that 150/105 carry $\alpha = -\frac{1}{2}$ is the same statement seen from the fit side, and (7) now supplies it exactly. Figure 1 (right) overlays the exact ratio on (7).

5 Numerical frontier

Two frontiers are reported, deliberately kept separate.

- The **assumption-free frontier**: the streamed engine (`graph.scc.sectors_union_find` over `core.cycle.succ`), which propagates exact $\mathbb{Z}[1/\sqrt{2}]$ amplitudes and uses no result of this report. Its cost grows by a factor ≈ 3 per site because the one-cycle support itself grows exponentially, so it is the binding runtime constraint: $N = 20$ already takes 65 minutes.
- The **reduction frontier**: Theorem 1 implemented as vectorised label propagation on the flip graph, with the label array (`uint32`, 4 bytes per basis state) the only object of size 2^N — pointer jumping is done in place and in chunks, and the size histogram is harvested chunkwise, so nothing else of that size is ever allocated. This is RAM-bound

rather than time-bound, and it reaches $N = 32$: $2^{32} = 4\,294\,967\,296$ basis states, 17 sectors, $D_{\max} = 1\,166\,803\,110$, in 815 s at a peak of 21.0 GB. That is the ceiling twice over — $N = 32$ is the largest index a `uint32` label can hold, and $N = 33$ would need a 64-bit label array of 68.7 GB.

Both agree with each other and with (6) at every N computed. Table 1 lists the units; run-times are wall-clock on the project machine (20 cores, 31 GB, jobs `niced` and run concurrently).

N	2^N	#sectors	$D_{\max}$	$D_{\max}/2^N$	engine	flip graph
14	16 384	8	6 435	0.3928	✓ 4.4 s	—
15	32 768	9	12 870	0.3928	✓ 14 s	—
16	65 536	9	24 310	0.3709	✓ 40 s	—
17	131 072	10	43 758	0.3338	✓ 126 s	—
18	262 144	10	92 378	0.3524	✓ 363 s	—
19	524 288	11	184 756	0.3524	✓ 1321 s	—
20	1 048 576	11	352 716	0.3364	✓ 1.1 h	—
21	2 097 152	12	646 646	0.3083	—	✓ 0.3 s
22	4 194 304	12	1 352 078	0.3224	—	✓ 0.4 s
23	8 388 608	13	2 704 156	0.3224	—	✓ 0.9 s
24	16 777 216	13	5 200 300	0.3100	—	✓ 1.8 s
25	33 554 432	14	9 657 700	0.2878	—	✓ 3.7 s
26	67 108 864	14	20 058 300	0.2989	—	✓ 8.9 s
27	134 217 728	15	40 116 600	0.2989	—	✓ 18 s
28	268 435 456	15	77 558 760	0.2889	—	✓ 38 s
29	536 870 912	16	145 422 675	0.2709	—	✓ 72 s
30	1 073 741 824	16	300 540 195	0.2799	—	✓ 154 s
31	2 147 483 648	17	601 080 390	0.2799	—	✓ 297 s
32	4 294 967 296	17	1 166 803 110	0.2717	—	✓ 815 s

Table 1: C150 `obc0`: exact sector data and the two numerical frontiers. ✓ means the computed size multiset equals the closed form (6) entry for entry; the time is that unit's wall clock; "—" means that route has not been run at that N (the engine is a factor ≈ 3 per site, so its units above $N = 20$ are hours to days each). The closed-form columns are exact for every N, computed or not.

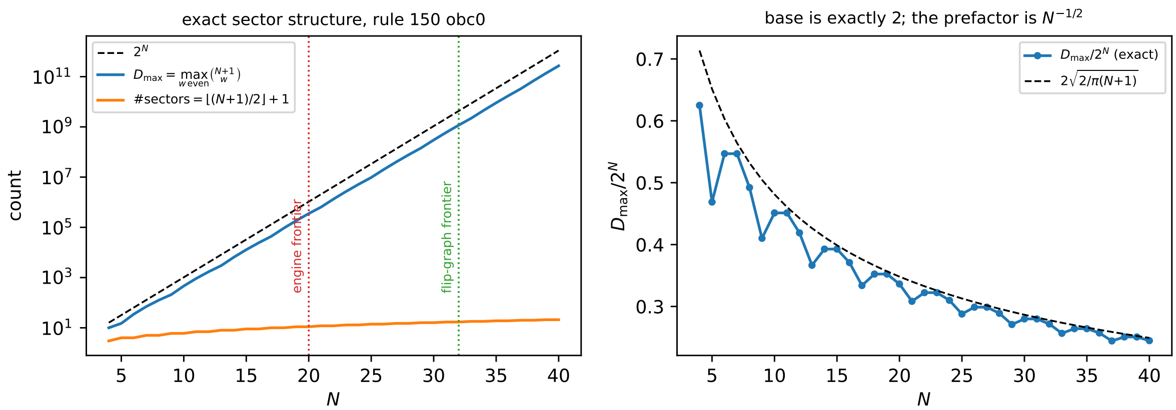

Figure 1: Left: the exact sector structure (6) with the two computed frontiers marked. Right: $D_{\max}/2^N$ against the asymptotic $2\sqrt{2/\pi(N+1)}$ of (7) — the base is 2 and the decay is the $N^{-1/2}$ prefactor that a pure-exponential fit misreads as 2.022.

6 What the sector does *not* determine

A sector is the minimal U-invariant subspace spanned by basis states: the *maximal kinematically reachable* set from any of its members, and nothing more (R1, §"What a sector is"). For C150 we can now quantify the gap exactly, because the shells are small enough to diagonalise. The dataset is every shell of every $N = 4 \dots 24$ with $|S_w| \leq 3000$: 70 shells, the largest of dimension 2380 ($N = 16$, $w = 4$). All numbers below come from U_w built column by column from the engine's exact amplitudes and verified orthogonal to 10^{-9}. The eigenvalue clustering tolerance is 10^{-9}, and it has margin in both directions: across the whole dataset the degeneracies of U_w are exact (largest gap treated as a degeneracy 8.8×10^{-15}, five orders below the tolerance) while the smallest gap treated as *distinct* is 4.8×10^{-6}, more than three orders above it.

An exponentially large stationary space. U does not merely fail to be ergodic inside a sector; it has a huge kernel of $U - \mathbb{1}$:

$$\boxed{\dim \mathrm{Fix}(U_w) \;=\; \binom{\lceil N/2 \rceil}{w/2}, \qquad \dim \mathrm{Fix}(U) \;=\; 2^{\lceil N/2 \rceil} \;=\; 2^{\#\{\text{even sites}\}}.} \tag{8}$$

The shell law is verified over the whole dataset — every shell of every $N = 4 \dots 24$ with $|S_w| \leq 3000$, 70 shells, no mismatch; the total is verified for $N = 3 \dots 12$ by two independent routes, the multiplicity of the eigenvalue $+1$ and $\dim \ker(U - \mathbb{1})$ by rank. The mechanism is visible in the half-layer structure: writing $U = L_B L_A$ with L_A, L_B the (Hermitian, involutive) half-layers, one finds *numerically for* $N = 3 \dots 11$ that

$$\mathrm{Fix}(U) \;=\; \mathrm{Fix}(L_A) \cap \mathrm{Fix}(L_B), \tag{9}$$

i.e. a state stationary under the cycle is already stationary under each half-layer separately, and each half-layer is a product of commuting single-qubit involutions (H or $\mathbb{1}$ per site of its sublattice), each of which contributes a one-dimensional local $+1$ eigenspace. The counting exponent $\lceil N/2 \rceil$ is the number of even sites, i.e. the width of the first half-layer. Only the $w = 0$ (and, for odd N, $w = N + 1$) members of $\mathrm{Fix}(U)$ are basis states; *all the rest are dark superpositions*, invisible to any computational-basis graph.

Consequence: an inner sector is never a Krylov space. For a basis state $|x\rangle$, $\dim \mathcal{K}(x)$ equals the number of distinct eigenvalues of U_w that $|x\rangle$ overlaps, so any degeneracy bounds it strictly below $|S_w|$. Over the whole dataset the dividing line is sharp and depends only on the sector dimension:

$$\begin{aligned} \text{spectrum of } U_w \text{ nondegenerate} \;&\iff\; |S_w| \leq N + 1 \\ &\iff\; w \in \{0,\, N+1\} \quad \text{or} \quad (N \text{ even and } w = N), \end{aligned} \tag{10}$$

i.e. exactly the extremal shells — the two frozen singletons and, for even N, the top shell of dimension $N + 1$, where (8) gives $\dim \mathrm{Fix}(U_w) = 1$ and no further multiplicity appears. For every one of the 59 *inner* shells ($|S_w| > N + 1$) the spectrum is degenerate and

$$\dim \mathcal{K}(x) \;<\; |S_w|, \qquad \frac{\dim \mathcal{K}(x)}{|S_w|} \in [0.395,\, 0.963], \tag{11}$$

the fraction decreasing with sector size (Table 2, Figure 2). This is the sharp form, for this rule, of the general caveat in R1: the graph gives $\dim \mathcal{K}(x) \leq |S_w|$, and away from the extremal shells the inequality is always strict. For odd N the reflection P commutes with U exactly ($[P, U] = 0$ to machine zero, 30 of the 70 shells) and reflection-symmetric seeds lose roughly half their Krylov space again ($N = 7$, $w = 2$: 13 instead of 25) — the symmetry-forced degeneracy anticipated in R1, here measured. For even N the brick-wall order maps the sublattices into each other and reflection acts instead as a time reversal, $PUP = U^{-1}$ exactly (the other 40 shells).

Monitored versus unmonitored, made numerical. Under monitoring the sector *is* filled uniformly, provably: $M_w = |U_w|^2$ is doubly stochastic, irreducible on S_w (Corollary 2) and aperiodic (by the R1 no-interference theorem $\langle x|U|x\rangle \neq 0$ for every x, so every node carries a self-loop), hence $M_w^t \to$ uniform on S_w. Without monitoring the long-time state is the diagonal ensemble $\bar\rho = \sum_\lambda P_\lambda |x\rangle\langle x| P_\lambda$, and it is far from uniform: over the 59 inner shells its effective dimension $d_{\text{eff}} = 1/\operatorname{tr}\bar\rho^2$ occupies between 6.6% and 50% of the sector — never more than half — and the most-occupied basis state carries 2.0 to 12.5 times the uniform weight (Table 2). So for C150 the two experiments genuinely disagree about the long-time state *inside* a sector, even though they agree exactly about which sector that state lives in, and (by R1) about populations after a single cycle.

N	w	$\lvert S_w\rvert$	#spec	dim Fix	$\binom{\lceil N/2\rceil}{w/2}$	$\dim\mathcal{K}/\lvert S_w\rvert$	$d_{\text{eff}}/\lvert S_w\rvert$	$\max p\cdot\lvert S_w\rvert$	reflection
8	2	36	33	4	4	0.917	0.388	2.57	$PUP = U^{-1}$
8	4	126	81	6	6	0.643	0.222	4.46	$PUP = U^{-1}$
8	6	84	65	4	4	0.774	0.292	3.42	$PUP = U^{-1}$
10	2	55	51	5	5	0.927	0.359	2.78	$PUP = U^{-1}$
10	4	330	211	10	10	0.639	0.156	6.33	$PUP = U^{-1}$
10	6	462	243	10	10	0.526	0.137	7.17	$PUP = U^{-1}$
10	8	165	131	5	5	0.794	0.241	4.12	$PUP = U^{-1}$
10	$10^\dagger$	11	11	1	1	1.000	0.793	1.26	$PUP = U^{-1}$
12	2	78	73	6	6	0.936	0.337	2.96	$PUP = U^{-1}$
12	4	715	473	15	15	0.662	0.123	8.07	$PUP = U^{-1}$
12	6	1 716	729	20	20	0.425	0.079	12.49	$PUP = U^{-1}$
12	8	1 287	665	15	15	0.517	0.096	10.15	$PUP = U^{-1}$
12	10	286	233	6	6	0.815	0.218	4.58	$PUP = U^{-1}$
12	$12^\dagger$	13	13	1	1	1.000	0.785	1.27	$PUP = U^{-1}$
14	2	105	99	7	7	0.943	0.316	3.13	$PUP = U^{-1}$
14	4	1 365	939	21	21	0.688	0.106	9.37	$PUP = U^{-1}$
14	12	455	379	7	7	0.833	0.201	4.98	$PUP = U^{-1}$
14	$14^\dagger$	15	15	1	1	1.000	0.779	1.28	$PUP = U^{-1}$
16	2	136	129	8	8	0.949	0.312	3.20	$PUP = U^{-1}$
16	4	2 380	1 697	28	28	0.713	0.093	10.75	$PUP = U^{-1}$
16	14	680	577	8	8	0.849	0.191	5.08	$PUP = U^{-1}$
16	$16^\dagger$	17	17	1	1	1.000	0.774	1.29	$PUP = U^{-1}$

Table 2: Wall shells of C150 `obc0`. #spec is the number of distinct eigenvalues of U_w; dim Fix its $+1$ multiplicity, beside the closed form (8); $\dim\mathcal{K}/|S_w|$ the largest Krylov fraction over the sampled seeds; $d_{\text{eff}}/|S_w|$ the smallest diagonal-ensemble fraction; $\max p\cdot|S_w|$ the peak long-time population in units of the uniform value (1 would be uniform, which is what monitoring gives exactly).

7 C150 is charge-resolved but interacting

The wall picture invites the guess that C150 is a free-fermion circuit. It is not, and the obstruction is exactly two signs.

Write the cycle on the $N+1$ bond modes. The gate at site i touches bonds i and $i+1$ only; it is the identity when both carry the same occupation, and on the one-wall subspace — basis ordered (wall at bond i, wall at bond $i+1$) — it is

$$M(c) \;=\; \frac{1}{\sqrt{2}}\begin{pmatrix} -(-1)^c & 1 \\ 1 & (-1)^c \end{pmatrix}, \qquad \det M(c) = -1, \tag{12}$$

where $c = x_{i-1} = \bigoplus_{k<i} b_k$ is the parity of the walls strictly to the left of bond i — a Jordan–Wigner string. This bond-space circuit reproduces the engine's U under the bijection of Proposition 1 to 1.1×10^{-16} (Table 3, second column), so (12) *is* C150: a hard-core quantum walk of domain walls with a Hadamard coin.

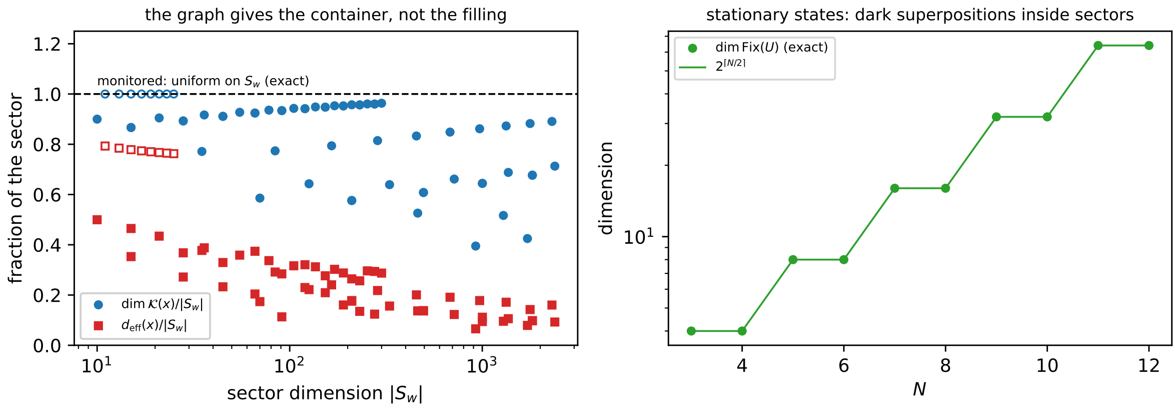

Figure 2: Left: the Krylov and diagonal-ensemble fractions of a sector against sector dimension; the dashed line is the monitored answer, which is exactly uniform. Filled markers are inner shells; open markers the extremal shells $|S_w| \le N+1$ — the only ones with a nondegenerate spectrum, and hence the only ones a basis state's Krylov space fills. Right: the stationary space $\dim \mathrm{Fix}(U) = 2^{\lceil N/2 \rceil}$, whose members other than the frozen basis states are dark superpositions.

Two features of (12) spoil gaussianity. A number-conserving Gaussian two-mode gate with single-particle block u must act on the doubly occupied state by $\det u$; here $\det M = -1$ while the gate acts by $+1$, which is a maximal nearest-neighbour interaction $\exp(i\pi n_i n_{i+1})$ between walls. And the $(-1)^c$ string multiplies the *density-difference* part of the coin, not the hop, so it is not absorbed by the Jordan–Wigner transform. The test is sharp: a number-conserving Gaussian circuit acts on the w-particle sector as the w-th antisymmetric power of its single-particle block, so its eigenphases are subset sums of the one-particle phases. Table 3 reports the largest failure of that additivity over $w = 2 \ldots 5$ for the four sign variants.

N	bond vs engine	rule 150	no string	det phase only	Gaussian ref.
5	1.1×10^{-16}	1.871	1.101	1.340	8.9×10^{-16}
6	1.1×10^{-16}	1.157	0.869	0.869	1.4×10^{-15}
7	1.1×10^{-16}	1.062	0.608	0.526	1.4×10^{-15}
8	1.1×10^{-16}	0.731	0.688	0.515	3.0×10^{-15}
9	1.1×10^{-16}	0.549	0.604	0.427	3.7×10^{-15}

Table 3: Additivity defect: the largest $|\arg(\lambda/\lambda')|$ in radians between the w-wall spectrum and the w-fold products of the one-wall eigenvalues, maximised over $w = 2 \ldots 5$ (compared as points on the unit circle, so the branch cut cannot masquerade as a defect). Column 2 is the bond representation checked against the engine. C150 (column 3) and each of the two single-defect models (columns 4, 5) are non-additive at 0.43–1.87 rad; removing *both* defects gives the Gaussian reference (column 6), additive to 4×10^{-15}.

The conclusion is a clean separation of the two questions. The *fragmentation* of C150 is entirely a $U(1)$ symmetry and is solved in closed form; the *dynamics within a sector* is that of interacting hard-core walls, and the interaction is not a perturbation — it is maximal. That is the honest reason a sector's spectral properties do not follow from (6), and why (8) had to be measured rather than read off the graph.

8 Open items

1. **Level statistics.** Deliberately not computed here. The spectra of every shell in Table 2 are reproducible in seconds from `quantum.rule150_spectra`, so a level-spacing analysis —

sector by sector, and separated by reflection parity for odd N — is immediately available; §7 predicts it will *not* look free, and (8) says the $+1$ eigenvalue must be excised first.

2. **A derivation of (8).** The law $\dim \mathrm{Fix}(U_w) = \binom{\lceil N/2 \rceil}{w/2}$ and the identity $\mathrm{Fix}(U) = \mathrm{Fix}(L_A) \cap \mathrm{Fix}(L_B)$ are exact numerical facts here, with the half-layer mechanism identified but not turned into a proof. The natural route is the local $+1$ eigenvector of each single-gate involution, together with the count of even sites; the w-resolved binomial suggests the dark states are labelled by choosing $w/2$ of the $\lceil N/2 \rceil$ even sites.

3. **The further exact multiplicities.** Beyond the $+1$ eigenvalue the spectrum carries exact multiplicities $2, 3, 4, 6, 10, 15$ (e.g. $N = 8$, $w = 4$: {1:48, 2:24, 3:8, 6:1}). These are not explained by reflection, which is abelian; some further commutant is present.

4. **The ring.** §4 solves `pbc` kinematically, but the quantum layer of §6–§7 is done for `obc0` only. On the ring the wall map is 2-to-1, so the bond representation acquires a $\mathbb{Z}_2$ gauge redundancy and (12) needs a flux sector; that is the next natural step.

5. **Higher N at the assumption-free frontier.** The streamed engine costs a factor ≈ 3 per site, so $N = 23$ is a $\approx$ 30-hour unit and sits outside the one-day budget used here. The reduction frontier is instead bounded by the label array: $N = 33$ needs `uint64` and 68.7 GB, so the next step there is a machine with more memory, or a compact relabelling (the components are few, so a two-pass scheme with a byte-wide label after the first sweep would halve the requirement).

9 Reproduce

```
# closed forms, wall bijection, charge conservation, engine cross-check
python -m qca_fragmentation.c150 --verify

# assumption-free frontier (one unit; N=22 takes ~10 h)
python -m qca_fragmentation.run_rule --rule 150 --N 21-22 --bc obc0

# reduction frontier (checkpointed to analytics/c150_frontier.jsonl)
python -m qca_fragmentation.c150 --frontier 21-31 --bc obc0

# sector spectra, Krylov dimensions, Fix(U), freeness test
python -m qca_fragmentation.scaling.c150_report --quantum
python -m qca_fragmentation.scaling.c150_report --tables --figures

# tests (126 of them)
cd code/qca_fragmentation/tests
python -m pytest test_flip_graph.py test_c150.py -q
```

Data: `results/150_obc0.jsonl` and `results/150_pbc.jsonl` (Tier 1a), `analytics/c150_frontier.jsonl` (reduction frontier, one JSON line per $(N, \texttt{bc})$ with the full size multiset and the closed-form check), `analytics/c150_quantum.json` (spectral portraits, $\mathrm{Fix}(U)$, freeness).